THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–5: **A Square and a Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 18	There are $2n = 18$ numbers between n^2 and $(n+1)^2$
2	**(b)** 8	$\frac{1}{a^{-n}} = a^n = 8$
3	**(d)** IC	I can be subtracted only from V and X; X only from L and C; C only from D and M; V, L, D are never subtracted
4	**(d)** 144°	One part $= 360 \div 10 = 36^\circ$
5	**(b)** Yes	alternate digit sums 9 and 9 differ by 0, a multiple of 11
6	**(b)** 29	$(n+1)^2 - n^2 = 2n + 1 = 29$
7	**(d)** 81	Even power gives a positive value; odd power gives a negative value
8	**(a)** 73	Multiply each digit by the power of the base and add
9	**(a)** 30	Opposite angles are equal, so the two expressions are equal
10	**(a)** Yes	even and digit sum 18 is a multiple of 3
11	**(c)** 36	$T_n = n(n+1)/2 = 36$
12	**(c)** 6×10^7	Place the decimal point after the first non-zero digit
13	**(c)** 322	Divide repeatedly by the base and read the remainders upwards
14	**(d)** 35°	A diagonal of a rhombus bisects the angle at the vertex
15	**(d)** 6174	Kaprekar routine: arrange digits in descending and ascending order and subtract

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 = 100$ (2).	2
17	$(2^3)^4 = 2^{12}$ (1); $2^{12} \times 2^{-5} = 2^7$ (1).	1+1
18	$3 \times 5^2 + 1 \times 5^1 + 4 \times 5^0$ (1); $= 84$ (1).	1+1
19	Total parts 12, one part $= 360 \div 12 = 30^\circ$ (1); smallest angle $= 30^\circ$ (1).	1+1
20	Sum of digits at odd places and even places: 13 and 13 (1); difference 0, so yes (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$T_5 = 15$, $T_6 = 21$ (1+1); sum $= 36 = 6^2$ (1).	1+1+1
22	(i) 2^3 (1); (ii) 2^{-6} (1); (iii) 2^7 (1).	1+1+1
23	$137 = 5 \times 27 + 2$; $27 = 5 \times 5 + 2$; $5 = 5 \times 1 + 0$; $1 = 5 \times 0 + 1$ (1); 1022_5 (1); check $1 \times 5^3 + 0 \times 5^2 + 2 \times 5^1 + 2 \times 5^0 = 137$ (1).	1+1+1
24	Let the angles be $1x, 2x, 3x, 4x$; sum $= 10x = 360$ (1); $x = 36$ (1); angles $36^\circ, 72^\circ, 108^\circ, 144^\circ$ (1).	1+1+1
	OR: By 8: Yes, last three digits 640 are divisible by 8 (1); by 11: Yes, alternate digit sums 9 and 9 differ by 0, a multiple of 11 (1); conclusion (1).	1+1+1
25	$22^2 = 484$, $120^2 = 14400$ (1); sum $= 14884$ (1); $122^2 = 14884$ equals the sum, so it is a triplet (1).	1+1+1
	OR: Add the exponents: $\left(\frac{2}{3}\right)^{-2}$ (1); $= \left(\frac{3}{2}\right)^2$ (1); value $\frac{9}{4}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	$1^3 + 12^3 = 1 + 1728 = 1729$ (2); $9^3 + 10^3 = 729 + 1000 = 1729$ (2); two different ways, so it is a taxicab number (1).	2+2+1
	OR: (i) $x + 3 = 11$, $x = 8$ (2); (ii) $4x = 20$, $x = 5$ (2); (iii) $x - 2 = -3$, $x = -1$ (1).	2+2+1
27	Column addition with carries gives 10100_2 (3); decimal check: $13 + 7 = 20$ and $10100_2 = 20$ (2).	3+2

	OR: (i) Draw a diagonal; it divides the quadrilateral into two triangles, each with angle sum 180° (3). (ii) $360 - (78 + 92 + 104) = 86^\circ$ (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–5: **A Square and a Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 49	$T_n + T_{n+1} = (n+1)^2 = 49$
2	**(b)** 0.00407	Move the decimal point according to the sign of the exponent
3	**(d)** 135	Base 60: value = 60 x first + second
4	**(b)** 50°	OA = OB, so triangle OAB is isosceles
5	**(d)** 39	Sum $= 3\times$ middle number
6	**(b)** 8, 15, 17	$2m = 8$, $m^2 - 1 = 15$, $m^2 + 1 = 17$
7	**(d)** 2^{-6}	$\frac{1}{2^6} = 2^{-6}$
8	**(b)** 72	Base 20: value = 20 x first + second
9	**(c)** 110°	Angle sum is 360°
10	**(d)** 80	Sum $= 5\times$ middle number
11	**(a)** 100	$1^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2 = 100$
12	**(d)** 7	The number being multiplied repeatedly is the base
13	**(a)** Hindu-Arabic system	Digits 0-9 with place value came from India and spread through Arabs
14	**(b)** 115°	Angles on the same non-parallel side add up to 180°
15	**(c)** 14	$(7+5)\times 2 - 10 = 2\times 7$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Take $2m = 6$, $m = 3$ (1); triplet $(6, 8, 10)$ (1).	1+1
17	$\left(\frac{3}{2}\right)^3$ (1) $= \frac{27}{8}$ (1).	1+1
18	$29 = 2\times 14 + 1$; $14 = 2\times 7 + 0$; $7 = 2\times 3 + 1$; $3 = 2\times 1 + 1$; $1 = 2\times 0 + 1$ (1); 11101_2 (1).	1+1
19	Adjacent angles are supplementary: $3x + 10 + 2x - 10 = 180$ gives $x = 36$ (1); $\angle A = 118^\circ$ (1).	1+1
20	Largest arrangement 6532, smallest 2356 (1); difference 4176 (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	XLIV = 44, LXVII = 67 (1); sum = 111 (1); CXI (1).	1+1+1
22	Diagonal AC bisects $\angle A$: $\angle OAB = 32^\circ$ (1); $\angle ABC = 180 - 64 = 116^\circ$ (1); $\angle OBA = 58^\circ$ (1).	1+1+1
23	Steps: 5321-1235=4086; 8640-0468=8172; 8721-1278=7443; 7443-3447=3996; 9963-3699=6264; 6642-2466=4176; 7641-1467=6174 (2); reaches 6174 after 7 steps (1).	2+1
24	$1156 = 2^2 \times 17^2$ (1); pairing (1); $\sqrt{1156} = 34$ (1).	1+1+1
	OR: (i) 5.9×10^6 (1); (ii) 7.4×10^{-4} (1); (iii) 8×10^4 (1).	1+1+1
25	$2\times 5^3 + 4\times 5^2 + 3\times 5^1 + 1\times 5^0$ (2); $= 366$ (1).	2+1
	OR: OA = OB so $\angle OAB = (180 - 70) \div 2 = 55^\circ$ (1); $\angle OAD = 90 - 55 = 35^\circ$ (1); $\angle AOD = 180 - 70 = 110^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Conditions of 5 and 9 on the stars (2); numbers: 4230, 4635 (2); verify one (1).	2+2+1
	OR: $13824 = 2^9 \times 3^3$ (1); triplets give $\sqrt[3]{13824} = 24$ (2); check $24^3 = 13824$ (2).	1+2+2
27	$600000 = 6\times 10^5$ (1); $2000 = 2\times 10^3$ (1); product 1.2×10^9 (2); quotient 3×10^2 (1).	1+1+2+1
	OR: $13 = 1101_2$ (2); weights used: 1, 4, 8 kg (2); each 1 in the binary number means that weight is used (1).	2+2+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–5: **A Square and a Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 24	There are $2n = 24$ numbers between n^2 and $(n+1)^2$
2	**(a)** 5	One lakh is 10^5
3	**(a)** VX	I can be subtracted only from V and X; X only from L and C; C only from D and M; V, L, D are never subtracted
4	**(d)** The diagonals bisect each other	Parallelogram diagonals bisect each other
5	**(b)** 1	Check the rule for 9
6	**(b)** 39	$(n+1)^2 - n^2 = 2n + 1 = 39$
7	**(b)** 9	$\frac{1}{a^{-n}} = a^n = 9$
8	**(c)** 125	Multiply each digit by the power of the base and add
9	**(d)** 144°	One part $= 360 \div 15 = 24^\circ$
10	**(a)** odd	Odd + odd = even, even + odd = odd
11	**(a)** 55	$T_n = n(n+1)/2 = 55$
12	**(a)** -32	Even power gives a positive value; odd power gives a negative value
13	**(b)** 226	Divide repeatedly by the base and read the remainders upwards
14	**(c)** 15	Opposite angles are equal, so the two expressions are equal
15	**(b)** Yes	alternate digit sums 11 and 11 differ by 0, a multiple of 11

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	The first odd number is $n^2 - n + 1 = 7$ (1); $7 + 9 + 11 = 27$ (1).	1+1
17	1.496×10^8 (2).	2
18	Babylonian: $3 \times 60 + 25 = 205$ (1); Maya: $4 \times 20 + 11 = 91$ (1).	1+1
19	Sum $= 360^\circ$ (1); it equals 360°, so it is possible (1).	1+1
20	$(13 + 6) \times 3 = 57$; minus 18 gives 39 (1); result $= 3 \times 13$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	47+74=121 (1); $121 = 11 \times 11$ (1); in general $10a + b + 10b + a = 11(a + b)$ (1).	1+1+1
22	$T_8 = 36$, $T_9 = 45$ (1+1); sum $= 81 = 9^2$ (1).	1+1+1
23	(i) 3^{-2} (1); (ii) 3^{-6} (1); (iii) 3^7 (1).	1+1+1
24	$214 = 8 \times 26 + 6$; $26 = 8 \times 3 + 2$; $3 = 8 \times 0 + 3$ (1); 326_8 (1); check $3 \times 8^2 + 2 \times 8^1 + 6 \times 8^0 = 214$ (1).	1+1+1
	OR: Let the angles be $2x, 3x, 4x, 6x$; sum $= 15x = 360$ (1); $x = 24$ (1); angles 48°, 72°, 96°, 144° (1).	1+1+1
25	By 8: Yes, last three digits 128 are divisible by 8 (1); by 11: Yes, alternate digit sums 16 and 5 differ by 11, a multiple of 11 (1); conclusion (1).	1+1+1
	OR: $26^2 = 676$, $168^2 = 28224$ (1); sum $= 28900$ (1); $170^2 = 28900$ equals the sum, so it is a triplet (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) They bisect each other (1) at right angles (1). (ii) $\angle OAB = 28^\circ$ (1); $\angle OBA = 90 - 28 = 62^\circ$ (1); $\angle ABC = 124^\circ$ (1).	1+1+1+1+1
	OR: $83 - 38 = 45 = 9 \times 5$ (2); in general $(10a + b) - (10b + a) = 9(a - b)$ (2); hence divisible by 9 (1).	2+2+1
27	$2^3 + 16^3 = 8 + 4096 = 4104$ (2); $9^3 + 15^3 = 729 + 3375 = 4104$ (2); two different ways, so it is a taxicab number (1).	2+2+1
	OR: (i) $x + 2 = 9$, $x = 7$ (2); (ii) $5x = 35$, $x = 7$ (2); (iii) $x - 4 = -1$, $x = 3$ (1).	2+2+1

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